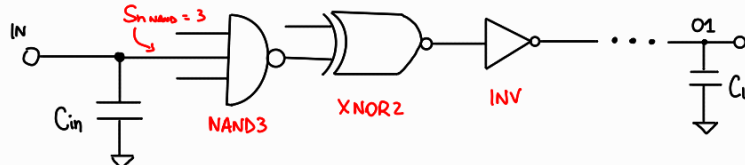


1)



$$\left. \begin{aligned} t_{po} &= R_{INV} C_{INV} \\ R_{INV} &= \frac{2F(V_T/V_{DD})}{\beta_n' S_{nINV} V_{DD}} \\ C_{INV} &= S_{nINV} (1+\epsilon) C_{H1} \end{aligned} \right\} t_{po} = \frac{2F}{\beta_n' S_{nINV} V_{DD}} \cdot S_{nINV} (1+\epsilon) C_{H1}$$

$$F\left(\alpha = \frac{V_T}{V_{DD}}\right) = \frac{1}{1-\alpha} \left[\frac{\alpha}{1-\alpha} + 0,5 \ln(18-20\alpha) \right] = 3,58$$

$$\epsilon = \frac{\beta_n'}{\beta_p'} = 1,5$$

$$C_{H1} = L^2 \mu_{IN} C_{OX} + 2L \mu_{IN} C_{GSO} \approx 0,23 \text{ pF}$$

$$t_{po} = \frac{2F}{\beta_n' S_{nINV} V_{DD}} \cdot S_{nINV} (1+\epsilon) C_{H1} = 21,21 \text{ pS}$$

$$\alpha_{NAND} = \frac{\epsilon}{n} = \frac{\epsilon}{3}$$

$$C_{IN} = \underbrace{S_{nNAND}}_3 \cdot \left(1 + \frac{\epsilon}{3}\right) C_{H1} = (3+\epsilon) C_{H1} \approx 0,86 \text{ fF}$$

$$p_{INV} = 0,9$$

$$g_{NAND} = \frac{n+\epsilon}{1+\epsilon} = \frac{3+\epsilon}{1+\epsilon} = 1,8$$

$$p_{NAND3} = n p_{INV} = 3 p_{INV} = 2,7$$

$$g_{XNOR} = 2$$

$$PD_{XNOR} = A \oplus B \longleftrightarrow (F = \overline{PD} = \overline{A \oplus B})$$

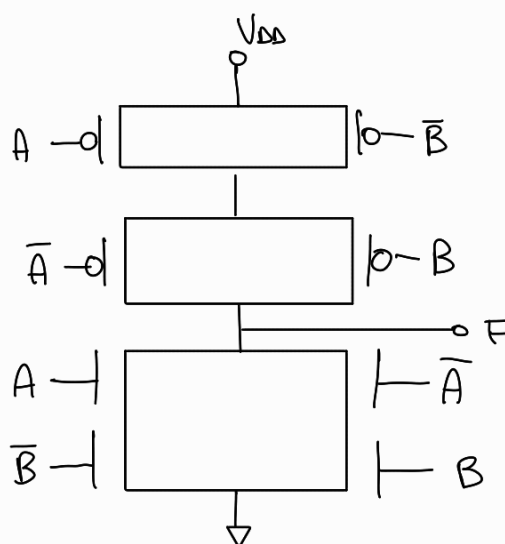
$$PD_{XNOR} = A\overline{B} + \overline{A}B$$

$$\alpha_{XNOR} = \epsilon$$

$$p_{XNOR} = \frac{C_P}{C_{INV}} \bigg|_{R_T = R_{INV}}$$

$$R_T = R_{INV}$$

$$\left. \begin{aligned} L \rightarrow S_{nEQ} &= S_{nINV} \\ S_{nEQ} &= \frac{S_n}{2} \end{aligned} \right\} S_n = 2 S_{nINV}$$



$$C_p = (2S_n + 2S_p) C_{p1} = 2S_n (1 + \epsilon) C_{p1}$$

$$= 4 S_n \text{inv} (1 + \epsilon) C_{p1}$$

$$P_{XNOR} = \frac{4 S_n \text{inv} (1 + \epsilon) \cancel{C_{p1}}}{S_n \text{inv} (1 + \epsilon) \cancel{C_{H1}}} = 4 p_{\text{inv}} = 3,6$$

$$g_{\text{inv}} = 1 \quad p_{\text{inv}} = 0,9$$

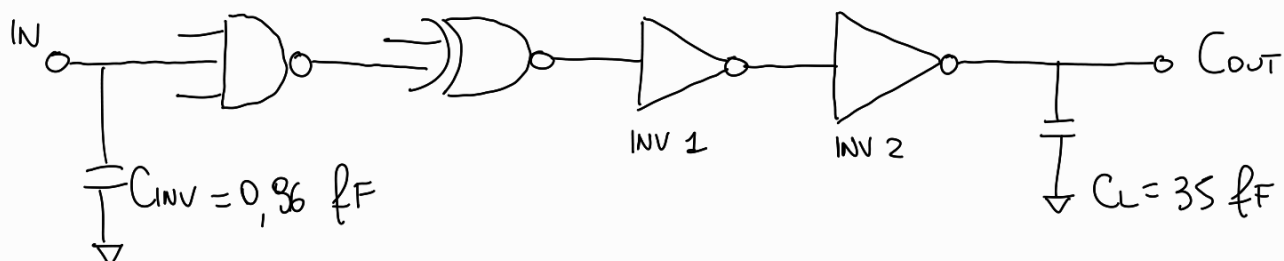
$$2) P \approx 0,71 p_{\text{inv}} + 2,82 = 3,459$$

$$F = \prod_{i=1}^N f_i = \prod_{i=1}^3 g_i b_i r_i = g_{\text{NAND3}} \cdot g_{\text{XNOR}} \cdot H = g_{\text{NAND3}} \cdot \frac{C_L}{C_{\text{IN}}}$$

$$= 131,25$$

$$\hat{N} = \frac{\ln F}{\ln 2} = 3,83 \longrightarrow N = 4 \quad (\text{deve essere intero!})$$

Aggiungo 1 invertitore al circuito:



$$\hat{f} = \sqrt[N]{F} \Big|_{N=\hat{N}} = \sqrt[4]{131,25} \approx 3,38$$

$$t_p = t_{p0} (4 \hat{f} + p_{\text{NAND3}} + p_{\text{XNOR}} + 2 p_{\text{inv}}) \approx 458 \text{ ps}$$

Per tutti gli stadi: $f_i = \hat{f} = g_i r_i = g_i \frac{C_{i+1}}{C_i}$

$$\hat{f} = g_{\text{INV2}} \cdot \frac{C_L}{C_{\text{INV2}}} \longrightarrow C_{\text{INV2}} = \frac{C_L}{\hat{f}} = 10,34 \text{ fF}$$

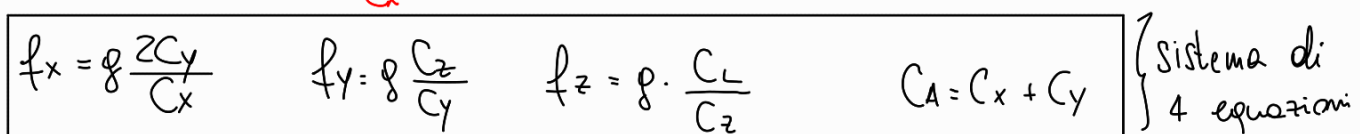
$$\hat{f} = g_{\text{INV1}} \cdot \frac{C_{\text{INV2}}}{C_{\text{INV1}}} \longrightarrow C_{\text{INV1}} = \frac{C_{\text{INV2}}}{\hat{f}} = 3,06 \text{ fF}$$

$$\hat{f} = g_{\text{XNOR}} \cdot \frac{C_{\text{INV1}}}{C_{\text{XNOR}}} \longrightarrow C_{\text{XNOR}} = g_{\text{XNOR}} \cdot \frac{C_{\text{INV1}}}{\hat{f}} = 1,81 \text{ fF}$$

$$\hat{f} = g_{\text{NAND3}} \cdot \frac{C_{\text{XNOR}}}{C_{\text{IN}}} \longrightarrow C_{\text{IN}} = g_{\text{NAND3}} \cdot \frac{C_{\text{XNOR}}}{\hat{f}} = 0,96 \text{ fF}$$

proporzionale a V_{Δ}^2

$$C_{INV1} = S_1(1+\varepsilon)C_{M1} = \frac{C_{INV2}}{\frac{1}{f}} \Rightarrow S_1 = 5,74$$



$$\Rightarrow C_x = \frac{2g C_y}{f_x} \quad \longrightarrow \quad C_x = \frac{2g^3 C_L}{f_x f_y f_z} \quad (1)$$

$$\Rightarrow C_y = g \frac{C_z}{f_y} \quad \left. \begin{array}{l} \Rightarrow C_z = g \frac{C_L}{f_z} \end{array} \right\} C_y = \frac{g^2}{f_y f_z} C_L \quad (2)$$

$$\Rightarrow C_A = C_x + C_y \quad (3)$$

Sostituisco (1) e (2) in (3):

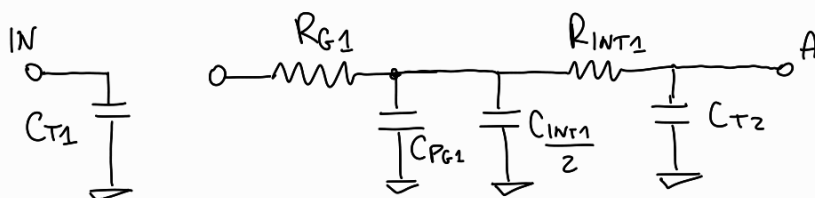
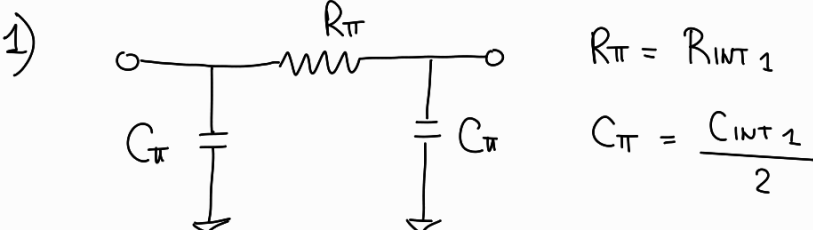
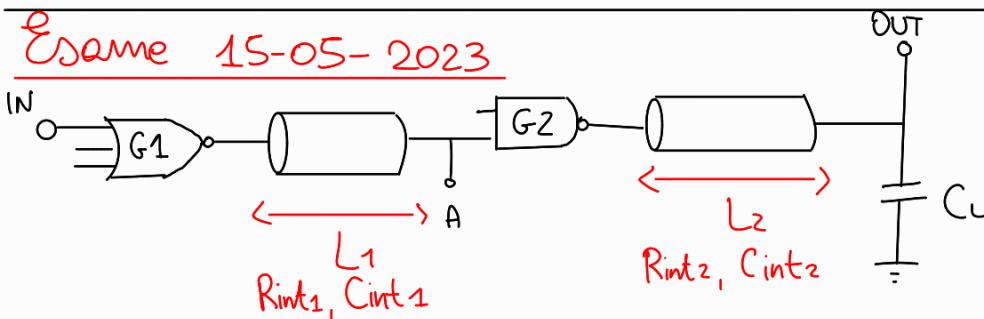
$$C_A = \frac{2g^3 C_L}{f_x f_y f_z} + \frac{g^2}{f_y f_z} C_L \quad | \quad f_x = f_y = f_z = f$$

$$C_A = \frac{2g^3 C_L}{f^3} + \frac{g^2 C_L}{f^2} \Rightarrow f^3 - 2g^3 \left(\frac{C_L}{C_A} \right) - f g^2 \left(\frac{C_L}{C_A} \right) = 0$$

Sappiamo che $\begin{cases} \frac{C_L}{C_A} = \frac{6fF}{2fF} = 3 \\ g = \frac{4}{3} \end{cases}$

Si può calcolare che $f = 3,14$

$$\frac{t_p}{t_{p0}} = \sum_{i=1}^3 f_i = 3f = 9,42 \longrightarrow t_p = 9,42 t_{p0}$$



$$R_{G1} = \frac{R_{T1}}{2,3}$$

$$t_{IN-A} = 2,3 \left[\left(C_{PG1} + \frac{C_{INT1}}{2} \right) R_{G1} + \left(\frac{C_{INT1}}{2} + C_{T2} \right) (R_{G1} + R_{INT1}) \right]$$

$$= 2,3 \left[C_{PG1} R_{G1} + \frac{C_{INT1}}{2} R_{G1} + \frac{C_{INT1}}{2} R_{G1} + \frac{C_{INT1}}{2} R_{INT1} + C_{T2} R_{G1} + C_{T2} R_{INT1} \right]$$

$$= 2,3 \left[(C_{T2} + C_{INT2}) R_{G1} + R_{G1} C_{PG1} + R_{INT1} \left(\frac{C_{INT1}}{2} + C_{T2} \right) \right]$$

$$= 2,3 \frac{t_{po}}{t_{po}} \left[(C_{T2} + C_{INT1}) \frac{R_{G1}}{2,3} + \frac{R_{G1}}{2,3} C_{PG1} + R_{INT1} \left(\frac{C_{INT1}}{2} + C_{T2} \right) \right]$$

$$= t_{po} \left[\underbrace{\frac{R_{T1} G_1}{t_{po}}}_{g_1} \underbrace{\left(\frac{C_{T2}}{C_{T1}} + \frac{C_{INT1}}{C_{T1}} \right)}_{h_1} + \underbrace{\frac{R_T C_{PG1}}{t_{po}}}_{p_1} + \underbrace{2,3 \frac{R_{INT1}}{t_{po}} \left(\frac{C_{INT1}}{2} + C_{T2} \right)}_{pw_1} \right]$$

$$= t_{po} \left[g_1 (h_1 + h_{w1}) + p_1 + pw_1 \right]$$

$$2) t_{IN \rightarrow A} = t_{po} \left[\underbrace{\frac{R_{T1} G_1}{t_{po}}}_{g_1} \underbrace{\left(\frac{C_{T2}}{C_{T1}} + \frac{C_{INT1}}{C_{T1}} \right)}_{h_1} + \underbrace{\frac{R_T C_{PG1}}{t_{po}}}_{p_1} + \underbrace{2,3 \frac{R_{INT1}}{t_{po}} \left(\frac{C_{INT1}}{2} + C_{T2} \right)}_{pw_1} \right]$$

$$t_{IN \rightarrow OUT} = t_{IN \rightarrow A} + t_{A \rightarrow OUT}$$

$$t_{A \rightarrow OUT} = t_{po} \left[\frac{R_{T2} C_L}{t_{po}} \left(\frac{C_L}{C_{T2}} + \frac{C_{INT2}}{C_{T2}} \right) + \frac{R_{T2} C_{PG2}}{t_{po}} + 2,3 \frac{R_{INT2}}{t_{po}} \left(\frac{C_{INT2}}{2} + C_L \right) \right]$$

$$\frac{C_{T2}}{C_{T1}} \triangleq h_1 \implies C_{T2} = h_1 C_{T1}$$

$$t_{A \rightarrow OUT} = t_{po} \left[\underbrace{\frac{R_{T2} G_2}{t_{po}}}_{g_2} \underbrace{\left(\frac{C_L}{h_1 C_{T1}} + \frac{C_{INT2}}{h_1 C_{T1}} \right)}_{h_2} + \underbrace{\frac{R_{T2} C_{PG2}}{t_{po}}}_{p_2} + \underbrace{2,3 \frac{R_{INT2}}{t_{po}} \left(\frac{C_{INT2}}{2} + C_L \right)}_{pw_2} \right]$$

$$t_{IN \rightarrow A} = t_{po} \left[\underbrace{\frac{R_{T1} G_1}{t_{po}}}_{g_1} \underbrace{\left(\frac{C_{T2}}{C_{T1}} + \frac{C_{INT1}}{C_{T1}} \right)}_{h_1} + \underbrace{\frac{R_T C_{PG1}}{t_{po}}}_{p_1} + \underbrace{2,3 \frac{R_{INT1}}{t_{po}} \left(\frac{C_{INT1}}{2} + C_{T2} \right)}_{pw_1} \right]$$

$$t_{IN \rightarrow OUT} = t_{IN \rightarrow A} + t_{A \rightarrow OUT}$$

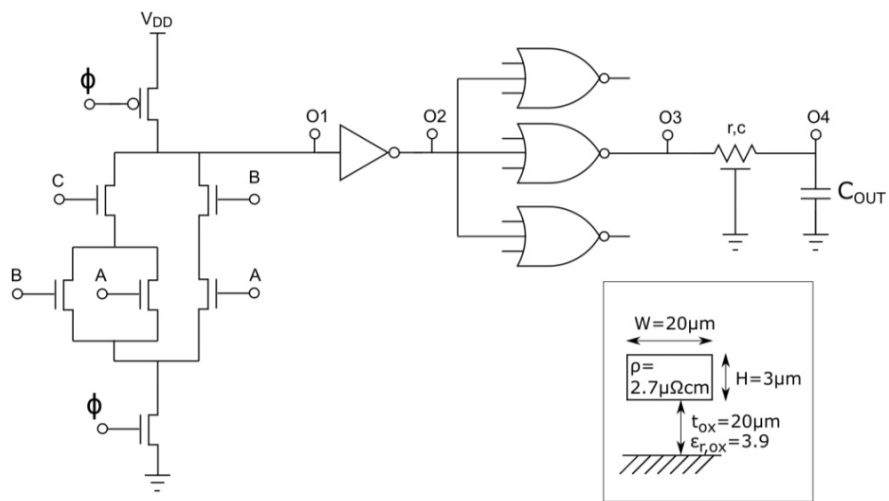
$$= t_{po} \left[g_1 h_1 + g_1 h_{w1} + p_1 + \frac{2,3 R_{INT1} C_{INT1}}{2 + p_0} + \frac{2,3 h_1 R_{INT1} C_{T1}}{t_{po}} + \frac{g_2}{h_1} \left(\frac{C_L}{C_{T1}} + \frac{C_{INT2}}{C_{T1}} \right) + p_2 + pw_2 \right]$$

$$\frac{\partial t_{IN-OUT}}{\partial h_1} = 0 \implies g_1 + \frac{2,3 R_{INT1} C_{T1}}{t_{po}} - \frac{g_2}{h_1^2 C_{T1}} (C_L + C_{INT2}) = 0$$

$$\implies \left(g_1 + \frac{2,3 R_{INT1} C_{T1}}{t_{po}} \right) h_1 = g_2 \left(\frac{C_L + C_{INT2}}{C_{T2}} \right)$$

$$\left(g_1 + \frac{2,3 R_{INT1} C_{T1}}{t_{po}} \right) \frac{C_{T2}}{C_{T1}} = g_2 \left(\frac{C_L + C_{INT2}}{C_{T2}} \right)$$

$$C_{T2} = \sqrt{\frac{C_{T1} (C_L + C_{INT2}) g_2}{g_1 + \frac{2,3 R_{INT1} C_{T1}}{t_{po}}}}$$



$$1) \varphi_A = \frac{R_T C_T}{R_{INV} C_{INV}} = \frac{C_T}{C_{INV}} \Big|_{R_T = R_{INV}}$$

$$S_{n\text{eq}} = \frac{S_n}{3} \Rightarrow R_T = R_{\text{inv}} \Rightarrow \frac{2F(V_T/V_{DD})}{\beta'_n S_{n\text{eq}} V_{DD}} = \frac{2F(V_T/V_{DD})}{\beta'_n S_{n\text{inv}} V_{DD}}$$

$$S_{neg} = \frac{S_n}{3} = S_n^{inv} \Rightarrow S_n = 35^{inv}$$

$$C_{inv} = S_{hinv} (1 + \varepsilon) C_{M1}$$

$$C_T = C_A = 2 S_m C_{H1} = 6 S_{minv} \cdot C_{H1}$$

$$Q_A = \frac{C_T}{C_{INV}} \bigg|_{R_T = R_{INV}} = \frac{GS_{minv} \cdot C_{H1}}{S_{minv} (1+\varepsilon) C_{H1}} = \frac{G}{1+\varepsilon} = 2$$

$$C_{pt} = (2S_n + S_p) C_{p1} = S_n (2 + \alpha) C_{p1}$$

$$t_1 = t_f \Big|_{\text{caso peggiore}} \Rightarrow S_p \beta'_p = \frac{S_m}{3} \beta'_m \Rightarrow \frac{S_p}{S_m} = \alpha = \frac{\varepsilon}{3}$$

$$C_{pt} = S_m (2 + \epsilon/3) C_{p1}$$

$$p = \frac{C_{pt}}{C_{inv}} \bigg|_{R_T = R_{inv}} = \frac{3 S_{minv} (2 + \varepsilon/3) C_{p1}}{S_{minv} (1 + \varepsilon) C_{H1}} = 3,2$$

$$2) R_T = \frac{2F (V_T/V_{DD})}{\beta'_m \frac{S_m}{3} V_{DD}} \quad R_{INV} = \frac{2F (V_T/V_{DD})}{\beta'_m V_{DD}}$$

$$R_T = \frac{R_{inv}}{2} \Rightarrow \frac{2F(V_T/V_{DD})}{\beta'_m \frac{S_m}{3} V_{DD}} = \frac{2F(V_T/V_{DD})}{2\beta'_m V_{DD}} \Rightarrow S_m = 6$$

$$t_{po} = R_{inv} C_{inv} = \frac{2F(V_T/V_{DD})}{\beta_{in} S_m V_{DD}} \cdot S_m (1+\varepsilon) C_{H1} = 8,94 \text{ ps}$$

$$F\left(\frac{V_T}{V_{DD}}\right) = F(r) = \frac{1}{1-r} \left[\frac{r}{1-r} + 0,5 \ln(18-20r) \right]$$

3) $S_m = 6$ per il gate dinamico

$$S_{m \text{ inv}} = 1$$

$$S_{m \text{ NOR}} = 6$$

$$C_{out} \ll C_{int}$$

$$\epsilon = 2$$

$$t_{A \rightarrow O4} = t_{A \rightarrow O1} + t_{O1 \rightarrow O2} + t_{O2 \rightarrow O4}$$

$$t_{A \rightarrow O1} = t_{po} \left(g_A \frac{C_{INV1}}{C_A} + p \right) = 33,078 \text{ ps}$$

$$t_{O1 \rightarrow O2} = t_{po} \left(\frac{3C_{NOR3}}{C_{INV}} + p_{INV} \right) = 386,208 \text{ ps}$$

$$C_{NOR3} = S_{m \text{ NOR}} \left(1 + \underbrace{\alpha_{NOR}}_{m\epsilon} \right) C_{H1} = 6 (1 + 3\epsilon) C_{H1} = 31,5 \text{ fF}$$

$$t_{O2 \rightarrow O4} = \pi C_{Len}^2 + 2,3 R_{OR} C_{Len} \quad (C_L \ll C_{int})$$

$$R_{OR} = \frac{R_{T \text{ NOR3}}}{2,3} = \frac{2F}{2,3} \frac{\beta_m \cdot 6 \cdot V_{DD}}{\beta_m \cdot 6 \cdot V_{DD}} = 288 \Omega$$

$$\pi = \frac{\rho}{H \cdot w} = \frac{2,7 \mu\Omega \cdot \text{cm}}{(20 \mu\text{m})(3 \mu\text{m})} = 4,5 \cdot 10^{-4} \frac{\Omega}{\mu\text{m}}$$

$$C = \epsilon_r \epsilon_0 \left(\frac{W}{t_{ox}} + \frac{2\pi}{\ln \left(\frac{4t_{ox} + H}{H} \right)} \right) \simeq 0,1 \text{ fF}/\mu\text{m}$$

$$t_{O2 \rightarrow O4} = 332,425 \text{ ps}$$

$$t_{A \rightarrow O4} = 751,7 \text{ ps}$$

4) Per il ritardo $A \rightarrow O2$

$$g_1 b_1 r_1 = g_2 b_2 r_2 \quad r_i = \frac{C_{i+1}}{C_i}$$

$$C_{INV} = \sqrt{\frac{3C_{NOR3} C_A}{g_1}} = 20,62 \text{ fF}$$

Per il ritardo $O2 \rightarrow O4$

$$t_{O2 \rightarrow O4} = \pi C_{Len}^2 + 2,3 R_{OR} C_{Len}$$

Ci chiediamo se:

$$2,3 R_{OR} C_{Len} > \pi C_{Len}^2 \cdot 10 \implies C_{Len} < \frac{2,3 R_{OR}}{10\pi}$$

$\underbrace{\quad}_{5000 \mu\text{m}} \quad \underbrace{\quad}_{147200 \mu\text{m}}$

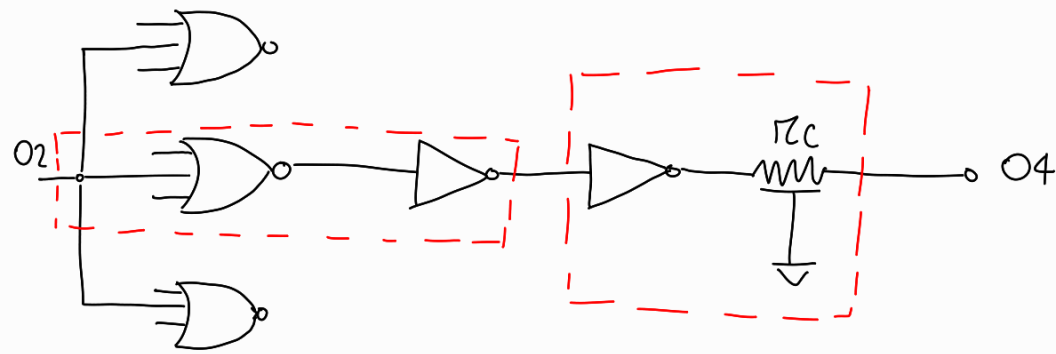
L'interconnessione può essere vista come una capacità.

$$P = 0,71 p_{inv} + 2,82 = 3,672$$

$$F = g_{NOR3} \cdot \frac{C_{int}}{C_{NOR3}} \simeq 37 \longrightarrow \hat{N} = \frac{\ln(F)}{\ln(P)} = 2,77$$

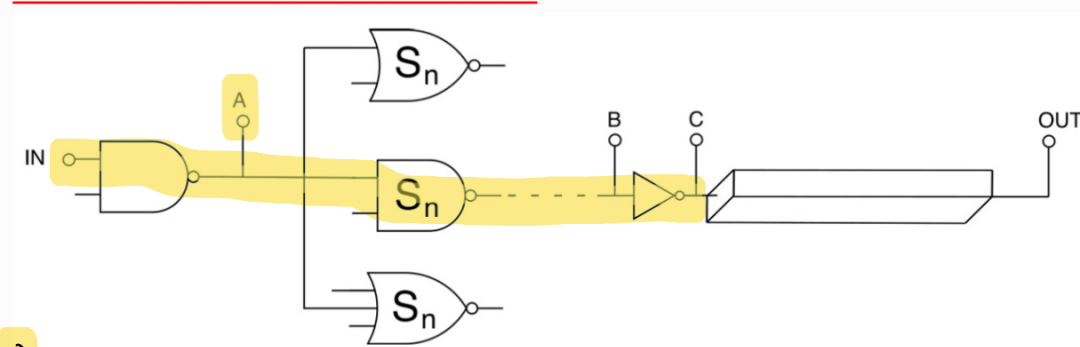
$N = 3 \longrightarrow$ dobbiamo aggiungere 2 invertitori tra l'uscita del NOR e l'interconnessione RC

$$\hat{f} = \sqrt[3]{F} = 3,33$$



$$t_{02 \rightarrow 04} = t_{p0} (2\hat{f} + p_{\text{NOR}} + p_{\text{INV}}) + \pi c L_{\text{em}}^2 + 2,3 R_{\text{oe}} c L_{\text{em}}$$

Esame del 16-05-2022



$$L = L_{\min}$$

$$t_{po} = 15 \text{ ps}$$

$$P_{inv} = 1,4$$

$$C_{M1} = 5fF$$

$nMOS \rightarrow 400 \mu A/V^2$
 $pMOS \rightarrow 200 \mu A/V^2$

$$L_{int} = 250 \mu m$$

1) $C_{WT} = C_{Len} = 2,5 \text{ pF}$

$$F = GBH$$

$$g_{\text{NAND2}} = \frac{2+\varepsilon}{1+\varepsilon} = 4/3 \Rightarrow G = (g_{\text{NAND2}})^2 = \frac{16}{9}$$

$$b_A = \frac{C_{\text{ON-PATH}} + C_{\text{OFF-PATH}}}{C_{\text{ON-PATH}}} = \frac{C_{\text{NAND2}} + (C_{\text{NOR2}} + C_{\text{NOR3}})}{C_{\text{NAND2}}}$$

$$\left. \begin{aligned} C_{\text{NAND } 2} &= S_m (1 + \varepsilon/2) C_{M1} \\ C_{\text{NOR } 2} &= S_m (1 + 2\varepsilon) C_{M1} \\ C_{\text{NOR } 3} &= S_m (1 + 3\varepsilon) C_{M1} \end{aligned} \right\} b_A = 7$$

$$H = \frac{C_{INT}}{C_{NAND2/N}} = \frac{C_{INT}}{4(1+\frac{\varepsilon}{2})C_{M1}} = \frac{2,5 \text{ pF}}{40 \text{ fF}} = 62,5$$

$$F = GBH \cong 778$$

$$2) \quad p = 0,71 \quad p_{inv} + 2,82 = 3,814$$

$$\hat{N} = \frac{\ln(F)}{\ln(p)} = 4,37 \longrightarrow N = 5 \longrightarrow \hat{f} = \sqrt[5]{F} = 3,786$$

$$t_{W \rightarrow B} = t_{PO} (\hat{4}t + 2\rho_{NAND2} + 2\rho_{INV}) \cong 353 \text{ ps}$$

Per ogni gate: $q_i b_i \pi_i = \hat{f}$

$$C_{INV3} \Rightarrow \frac{C_u}{C_{INV3}} = \hat{f} \Rightarrow C_{INV3} = \frac{C_u}{\hat{f}} = 660 \text{ fF}$$

$$C_{INV2} \Rightarrow \frac{C_{INV3}}{\hat{f}} = 174,4 \text{ fF}$$

$$C_{INV1} \Rightarrow \frac{C_{INV2}}{\hat{f}} = 46 \text{ fF}$$

$$C_{NAND2,2} = g_{NAND2} \cdot \frac{C_{INV2}}{C_{NAND2,2}} = \hat{f} \Rightarrow C_{NAND2,2} = \frac{g_{NAND2} \cdot C_{INV2}}{\hat{f}} = 16 \text{ fF}$$

$$C_{NAND2,1} = \frac{b_A \cdot g_{NAND2} \cdot C_{NAND2,2}}{\hat{f}} = 40 \text{ fF}$$

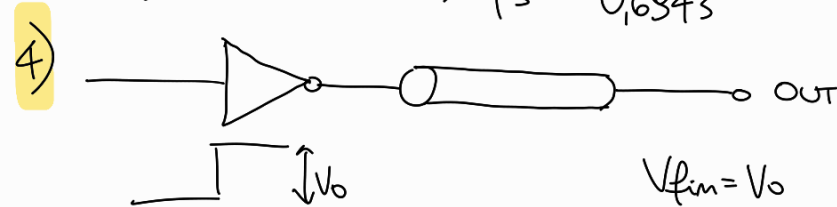
$$3) R_{DR} = \frac{R_{INV3}}{2,3}$$

$$g = \frac{R_T G}{R_{INV} C_{INV1}} = \frac{R_T G}{t_{po}} \Rightarrow R_T = \frac{t_{po} \cdot g}{C} \rightarrow R_T = R_{INV3} = 22,73 \text{ S}$$

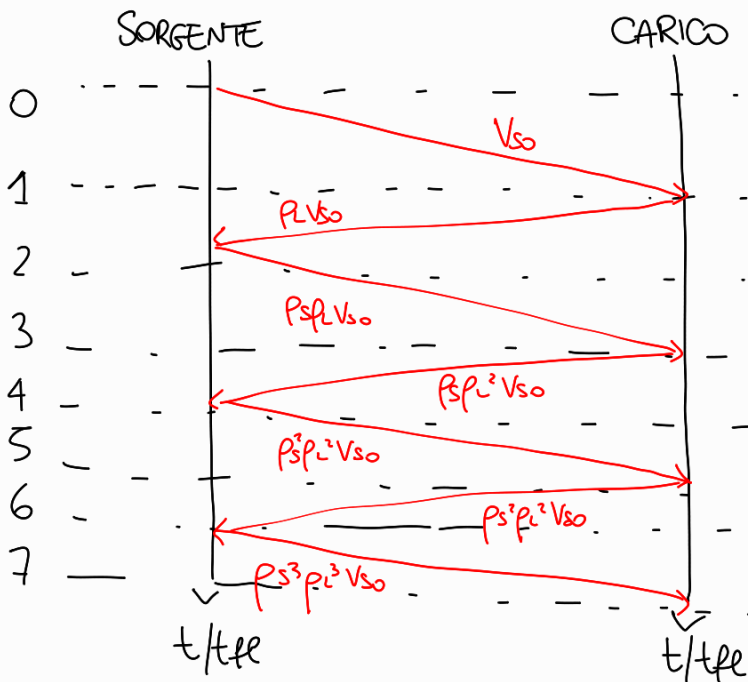
$$R_{DR} = \frac{R_{INV3}}{2,3} = 9,88 \text{ S}$$

$$p_s = \frac{R_s - Z_0}{R_s + Z_0} = \frac{R_{DR} - Z_0}{R_{DR} + Z_0} \quad p_L = \frac{Z_L - Z_0}{Z_L + Z_0} = 1$$

$$Z_0 = \sqrt{\frac{L}{C}} \approx 55 \text{ S} \Rightarrow p_s = -0,6843$$



$$V_{so} = \frac{Z_0}{Z_0 + R_{DR}} V_0 = 0,8477 V_0$$



$$V_L = (1 + p_L) V_{so} = 1,6854 V_0$$

$$V_L(3 + t_{fl}) = (1 + p_L) p_s p_L V_{so}$$

$$V_L(3 + t_{fl}) = V_L(t_{fl}) + V_L(3 + t_{fl}) = 0,5183 V_0$$

$$V_L(5 + t_{fl}) = (1 + p_L) p_s^2 p_L^2 V_{so}$$

$$V_L(5 + t_{fl}) = V_L(3 + t_{fl}) + V_L(5 + t_{fl}) = 1,3356 V_0$$

$$V_L(7 + t_{fl}) = (1 + p_L) p_s^3 p_L^3 V_{so}$$

$$V_L(7 + t_{fl}) = V_L(5 + t_{fl}) + V_L(7 + t_{fl}) = 0,7681 V_0$$

Servono 7 tfl per avere $V_{fin} \in [0,7 V_0, 1,3 V_0]$

$$t_{fl} = \frac{L_{en}}{V_e} \Rightarrow t_{totl} = \frac{7 L_{en}}{V_e}$$

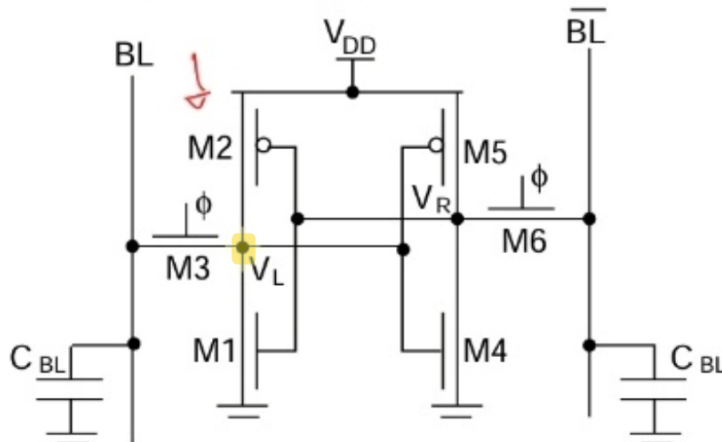
Prova Scritta di Complementi di Elettronica Digitale

20 Settembre 2010

Sia $V_{DD}=1.5V$ e si assumano i seguenti parametri tecnologici per i MOSFET:

Parametro	n-MOSFET	p-MOSFET
$V_{TO}[V]$	0.35	-0.35
$\beta'[\mu A/V^2]$	120	80
$\gamma[V^{1/2}]$	0.0	0.0
$\lambda[V^{-1}]$	0.0	0.0
$L_{MIN}[\mu m]$	0.13	0.13

Se non diversamente specificato si assuma per tutti i transistori $L=L_{MIN}$.

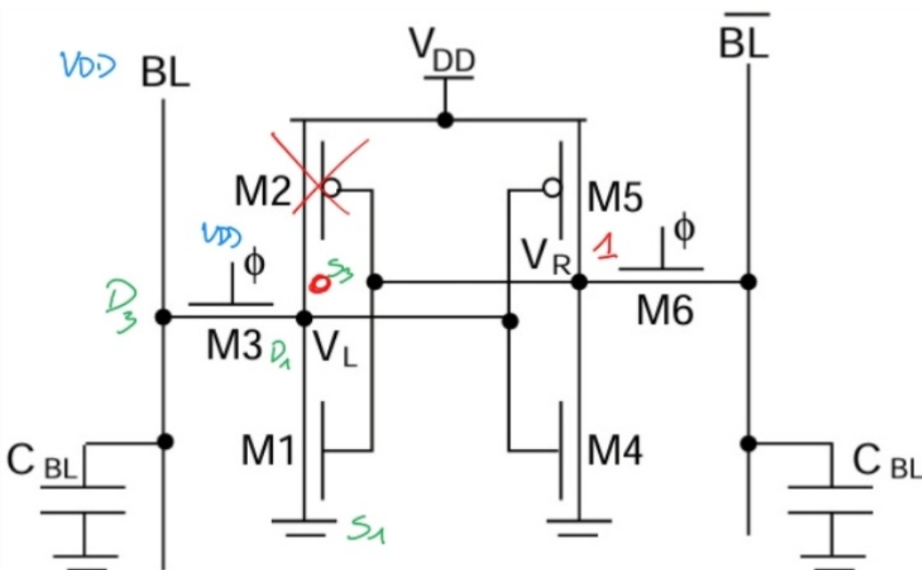


La figura mostra una cella di memoria RAM statica a sei transistori e la capacità della *bit-line* vale $C_{BL}=0.1pF$. In relazione a tale circuito si chiede di rispondere ai seguenti quesiti.

- 1) (12 Punti). Si consideri l'operazione di lettura in cui le due *bit-line* sono precaricate a V_{DD} ed il *bit* memorizzato è uno zero. Affinché la lettura non deteriori il dato memorizzato è necessario che l'aumento di tensione sul nodo V_L sia sufficientemente bassa. Questa specifica può essere imposta in modo approssimato valutando il valore V_{Lst} in condizioni stazionarie, cioè il valore che V_L raggiungerebbe se il transitorio di lettura fosse infinitamente lungo.

Detti S_1 ed S_3 i dimensionamenti di M1 ed M3, rispettivamente, si determini il rapporto $C_R=S_1/S_3$ che garantisce un V_{Lst} minore di 0.2V

Si supponga adesso di avere $S_3=1$ e $S_1=C_R S_3$ e si calcoli il tempo necessario affinché durante la lettura la tensione sulla *bit-line* si riduca di 100mV rispetto a V_{DD} .



M_1, M_3 accesi

Per M3:

$$V_{GS3} = V_{DD} - V_L \quad \left| \quad M_3 \text{ è saturo} \right.$$

$$V_{DS3} = V_{DD} - V_L$$

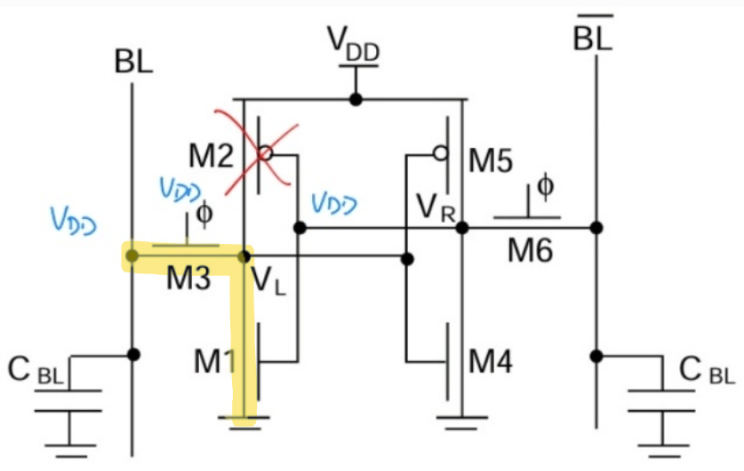
Per M1:

$$V_{GS1} = V_{DD}$$

$$V_{DS1} = V_{Lst} = 0.2V \quad \left| \quad M_1 \text{ è triodo} \right.$$

$$I_3 = I_1 : \frac{S_3 \beta'_m}{2} (V_{DD} - V_{Lst} - V_T)^2 = S_1 \beta'_m \left[(V_{DD} - V_T) V_{Lst} - \frac{V_{Lst}^2}{2} \right]$$

$$C_R = \frac{S_1}{S_3} = \frac{(V_{DD} - V_{Lst} - V_T)^2}{2} \cdot \frac{1}{(V_{DD} - V_T) V_{Lst} - V_{Lst}^2/2} \quad \text{per } V_{Lst} = 0,2V = 2,15$$



$$Seq_{3,1} = \frac{S_3 S_1}{S_3 + S_1} \simeq 0,68$$

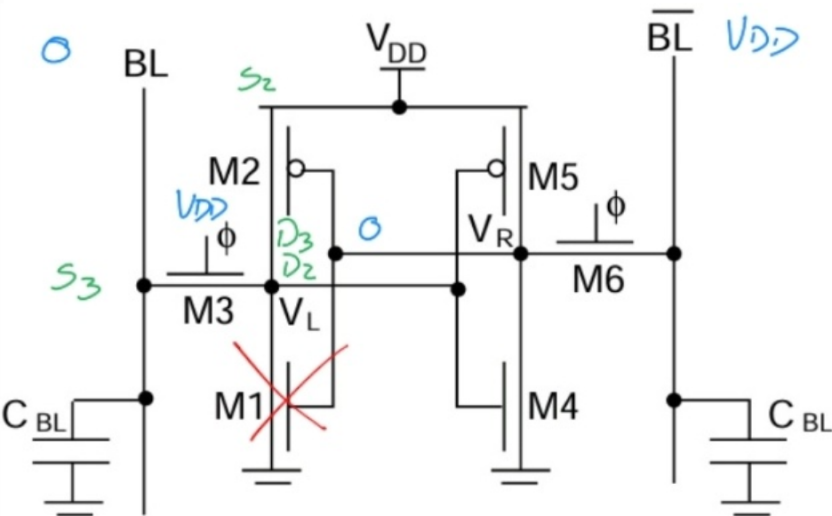
$V_{DS} > V_{GS} - V_T$ per tutto il transitorio di scarica

M_{eq} è saturato

$$V_{BL}(t) = V_{DD} - \frac{S_1 \beta'_m}{2} \frac{(V_{DD} - V_T)^2}{C_{BL}} \cdot t$$

$$V_{DD} - V_{BL}(t_f) = \Delta V = 100 \text{ mV} \implies t_f = \frac{2 C_{BL} \Delta V}{S_1 \beta'_m (V_{DD} - V_T)^2} = 185 \text{ ps}$$

2) (6 Punti). Si consideri ora l'operazione di scrittura di uno zero supponendo che il *bit* memorizzato sia un uno. Anche in questo caso si chiede di studiare la scrittura usando la tensione stazionaria V_{Lst} e, in particolare, di determinare il valore di S_2 che garantisce $V_{Lst} = 0.5V_{DD}$.



M_2, M_3 accesi

$$M_3: V_{GS3} = V_{DD} - 0 = V_{DD}$$

$$V_{DS3} = V_{LST} - 0 = V_{LST} = \frac{V_{DD}}{2}$$

$$V_{DS3} < V_{GS3} - V_T$$

M_3 è triodo

$$M_2: V_{GS2} = V_{DD}$$

$$V_{SD2} = V_{DD} - V_{LST} = V_{DD} - \frac{V_{DD}}{2} = \frac{V_{DD}}{2}$$

$$V_{SD2} < V_{GS} - |V_T| \longrightarrow M_2 \text{ è triodo}$$

$$I_3 = I_2 : \beta'_m S_3 \left[(V_{DD} - V_T) \frac{V_{DD}}{2} - \frac{(V_{DD}/2)^2}{2} \right] = \beta'_p S_2 \left[(V_{DD} - |V_T|) \frac{V_{DD}}{2} - \frac{(V_{DD}/2)^2}{2} \right]$$

$$S_2 = \frac{\beta'_m}{\beta'_p} S_3 = 1,5$$

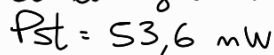
Si supponga che la cella sia simmetrica e quindi si abbia $S_1 = S_4$, $S_2 = S_5$ ed $S_3 = S_6$.

3) (8 Punti). Si calcoli la potenza statica P_{st} assorbita da una cella di memoria sapendo che la corrente I_{off} di un dispositivo avente $|V_{GS}| < |V_T|$ può essere approssimativamente espressa come:

$$I_{off} = S I_{sbth} \exp\left(-\frac{|V_{GS} - V_T|}{m_s V_{th}}\right)$$

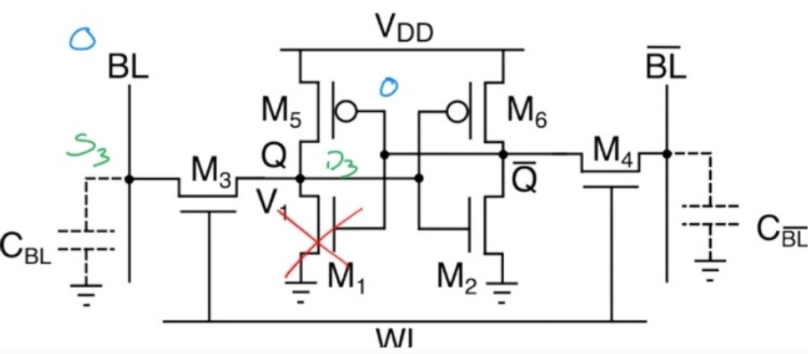
dove S è il dimensionamento del transistor, $I_{sbth} = 100 \mu A$ e $m_s = 1.25$ sono parametri tecnologici, mentre $V_{th} = 26 mV$ è la tensione termica (a temperatura ambiente).

Supponendo inoltre che, a causa delle variabilità nel processo di fabbricazione, i transistori della cella di memoria subiscano una riduzione in valore assoluto della soglia pari a $\Delta V_T = 50 mV$ (identica per tutti i transistori), si calcoli il nuovo valore della potenza statica P_{st} .



V_1 $\begin{cases} \rightarrow 2V \\ \rightarrow 0.2V \end{cases}$ $\rightarrow$ ESCLUSO PERCHÉ NON HA SENSO

2. (7 punti) Si calcoli il minimo valore del potenziale V_1 al drain di M_1 (si veda la figura) durante la scrittura di uno zero ($Q=0$) assumendo che il dato immagazzinato sia $Q=1$. Per fare ciò si assuma che lo switching della cella non sia iniziato e quindi che il potenziale ai gate dei transistor rimanga immutato e pari al caso con $Q=1$.



M_3 e M_5 accesi

$$V_{GS3} = V_{DD}$$

$$V_{DS3} = V_1$$

$$V_{DS3} < V_{GS3} - V_T$$

M_3 è triodo

$$M_5: V_{SG5} = V_{DD}$$

$$V_{SD5} = V_{DD} - V_1$$

Suppongo che V_1 a fine transitorio sia tale per cui:

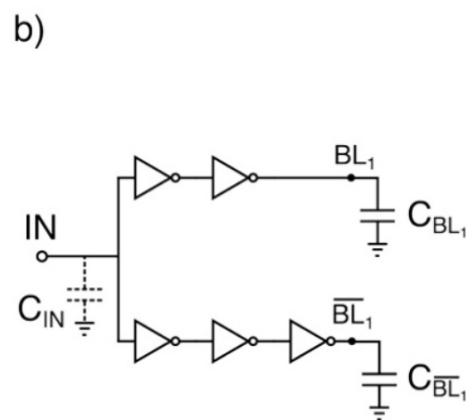
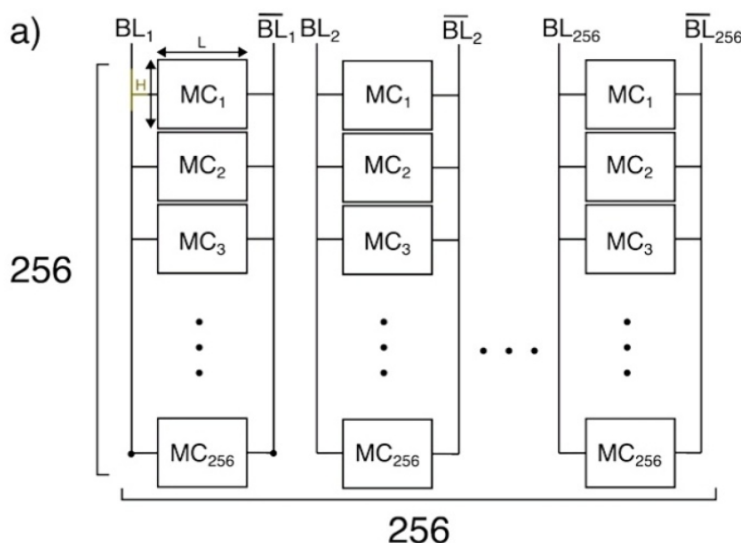
$$V_{DD} - V_1 = V_{SD5} > V_{SG5} - |V_T| = V_{DD} - |V_T| \quad (\text{DA VERIFICARE})$$

Ipotesi: M_5 è saturo

$$I_3 = I_5: S_3 \beta'_m \left[(V_{DD} - V_T) V_1 - \frac{V_1^2}{2} \right] = \frac{S_5 \beta'_p}{2} (V_{DD} - |V_T|)^2$$

$$V_1 \begin{cases} \rightarrow 1.96 \text{ V} \\ \rightarrow 0.23 \text{ V} \end{cases}$$

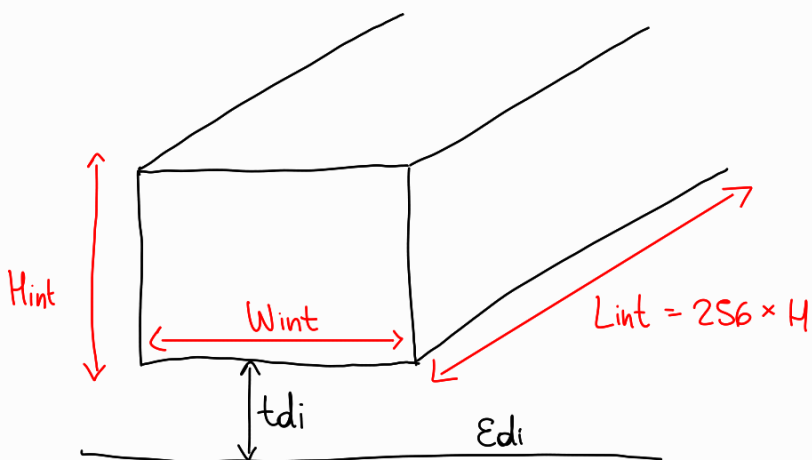
$$\begin{aligned} V_{SD5} &= V_{DD} - 0.23 \\ V_{SG5} - |V_T| &= V_{DD} - 0.4 \end{aligned} \quad \left| \begin{aligned} V_{SD5} &> V_{SG5} - |V_T| \\ (M_5 \text{ saturo}) \end{aligned} \right.$$



Si consideri una memoria SRAM costituita da 256×256 celle 6T come quella del punto precedente. Ogni cella ha larghezza $L=2\mu\text{m}$ e altezza $H=1.5\mu\text{m}$. Tutti i transistor hanno lunghezza di canale $L_G=0.1\mu\text{m}$ e un rapporto W/L come al punto precedente. La bitline realizzata sul metal M2 ha una larghezza $W_{int}=0.2\mu\text{m}$ e le capacità sono $c_{pp}=80\text{ aF}/\mu\text{m}^2$ e $c_{fr}=90\text{ aF}/\mu\text{m}/\text{lato}$. Le capacità di diffusione per unità di larghezza sono pari a $C_D = C_S = 1\text{ fF}/\mu\text{m}$.

1. (5 punti) Si calcoli la capacità di carico di ognuna delle bitline ($BL_1, \overline{BL}_1, \dots, BL_{256}, \overline{BL}_{256}$) del banco di memoria illustrato in figura a).

$$C_{pp} = 256 \times H \times W_{int} \times C_{pp}$$

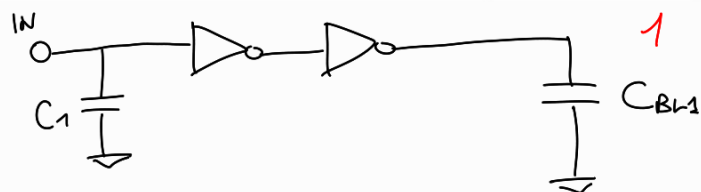
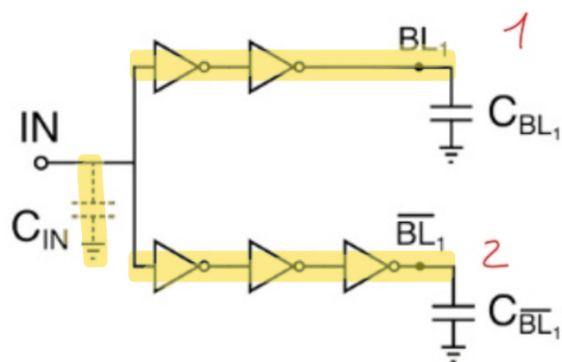


$$C_{fr} = 2 \times 256 \times H \times c_{fr}$$

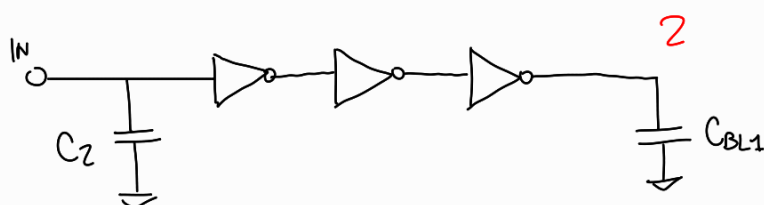
$$C_{diff} = 256 \times W_3 \times C_D = 256 \times \underbrace{S_3 \times L_6}_{W_3} \times C_D$$

$$C_{BL} = C_{pp} + C_{fr} + C_{diff} = 126,5 \text{ fF}$$

2. (11 punti) Si consideri il circuito di pilotaggio di figura b). Ogni coppia di bitline aventi capacità totale $C_{BL,1}$ e $C_{\overline{BL},1}$, sono pilotate da degli invertitori: 2 per BL_1 e 3 per $\overline{BL}_1$. Assumendo che la capacità che un eventuale driver vede al nodo di ingresso IN di figura b) sia un parametro di progetto (non può essere variata a piacere), utilizzando quanto appreso dalla metodologia del logical effort e fissato il numero di invertitori come da figura, si scriva l'espressione algebrica che consente di minimizzare il tempo $t_{IN \rightarrow BL_1}$ e $t_{IN \rightarrow \overline{BL}_1}$ e, in particolare, $t_{IN \rightarrow BL_1} = t_{IN \rightarrow \overline{BL}_1}$ e quindi calcolare i dimensionamenti dei transistori degli invertitori.



$$C_1 + C_2 = C_{IN}$$



Tutti gli stadi del ramo 1 devono avere stage effort $f_i = f_1$

Tutti gli stadi del ramo 2 devono avere stage effort $f_i = f_2$

$$f_1 = \sqrt{\frac{C_{BL}}{C_{IN1}}} \Rightarrow C_{IN1} = \frac{C_{BL}}{f_1^2} \quad \left| \begin{array}{l} t_{IN} \rightarrow B_L = t_{IN} \rightarrow \overline{B_L} \\ 2f_1 + 2p_{inv} = 3f_2 + 3p_{inv} \\ f_1 = \frac{3f_2 + p_{inv}}{2} \end{array} \right.$$

$$f_2 = \sqrt[3]{\frac{C_{BL}}{C_{IN2}}} \Rightarrow C_{IN2} = \frac{C_{BL}}{f_2^3}$$

$$\left\{ \begin{array}{l} C_{IN1} = \frac{C_{BL}}{f_1^2} \\ C_{IN2} = \frac{C_{BL}}{f_2^3} \\ f_1 = \frac{3f_2 + p_{inv}}{2} \\ C_{IN} = C_{IN1} + C_{IN2} \end{array} \right.$$

$\Rightarrow$

$$C_{IN} f_2^3 (3f_2 + p_{inv})^2 - 4C_{BL} f_2^3 - C_{BL} (3f_2 + p_{inv})^2 = 0$$